

# ZHIYUAN ZHU

[Email](#)[Website](#)[Google Scholar](#)[GitHub](#)

## EDUCATION

### ETH Zurich

M.Sc. in Robotics, Systems and Control (GPA: 6.0/6.0)

Zurich, Switzerland

Sep. 2024 – present

### Shanghai Jiao Tong University

B.Eng. in Automation (GPA: 3.93/4.3, rank 2/86)

Shanghai, China

Sep. 2020 – Jun. 2024

## PUBLICATIONS

- [1] **Zhiyuan Zhu**, Jiaxu Xing, Yunfan Ren, Ismail Geles, Yifan Zhai, Rudolf Reiter, Davide Scaramuzza. *Hybrid Neural Dynamics for Online Policy Adaptation in Agile Flight*. Under review, 2026.
- [2] Jiaxu Xing, **Zhiyuan Zhu**, Yunfan Ren, Ismail Geles, Yifan Zhai, Rudolf Reiter, Davide Scaramuzza. *Continual Robot Policy Learning via Variational Neural Dynamics*. arXiv preprint, 2026. [arXiv]
- [3] Yunfan Ren, **Zhiyuan Zhu**, Jiaxu Xing, Davide Scaramuzza. *Learning Agile Quadrotor Flight in the Real World*. Robotics: Science and Systems (RSS), 2026. [arXiv]
- [4] **Zhiyuan Zhu**, Yuanlong Li, Zongli Lin. *Global Practical Stabilization of the Double Integrator System with an Imperfect Sensor and Subject to a Bounded Disturbance*. Control Theory and Technology, 22(3):468–478, 2024. [DOI]
- [5] **Zhiyuan Zhu**, Yuanlong Li, Zongli Lin. *Global Practical Stabilization of the Double Integrator System in the Presence of Output Saturation and a Bounded Disturbance*. IFAC World Congress, Yokohama, Japan, 2023. [DOI]

## RESEARCH EXPERIENCE

### University of California, Berkeley

Master's Thesis, advised by Prof. Jitendra Malik

Berkeley, CA, USA

Aug. 2026 – present

- Developing an agentic sim-to-real transfer pipeline that probes a robot in simulation and in the real world, diagnoses the physical cause of the sim-to-real gap, and obtains a better-performing policy in the real world.

### ORCA Lab, Tongji University

Research Assistant, advised by Prof. Yunfan Ren

Shanghai, China

May 2026 – Aug. 2026

- Developed a JEPa-style world model that lets a quadrotor dodge thrown objects of unknown size and shape using a monocular camera via MPPI and a learning-based policy, transferring zero-shot to real hardware.

### Robotics and Perception Group, University of Zurich

Semester Thesis, advised by Prof. Davide Scaramuzza

Zurich, Switzerland

Sep. 2025 – May 2026

- Trained agile-flight policies for quadrotors with differentiable simulation, converging faster and with much higher sample efficiency than model-free RL (PPO), and deployed them on real hardware [3].
- Developed a hybrid neural dynamics model from real-world flight data and an online policy-adaptation framework that refines the policy in real time, keeping trajectory tracking robust under various real-world disturbances [1].

### Shanghai Jiao Tong University

Research Assistant, advised by Prof. Yuanlong Li and Prof. Zongli Lin

Shanghai, China

Jul. 2022 – Jun. 2024

- Designed a low-gain linear dynamic output feedback law that achieves global practical stabilization of the double integrator under output saturation and a bounded disturbance, verified in Simulink [5].
- Extended the design to imperfect sensors with both output dead zone and saturation, with a rigorous proof of global practical stability [4].

## HONORS & AWARDS

|           |                                                                                    |
|-----------|------------------------------------------------------------------------------------|
| 2024      | Zhiyuan Honors Bachelor's Degree, Shanghai Jiao Tong University                    |
| 2020–2024 | Zhiyuan Honors Scholarship, Shanghai Jiao Tong University (top 5%)                 |
| 2022      | Huawei Scholarship                                                                 |
| 2022      | Undergraduate Merit Scholarship (B Level), Shanghai Jiao Tong University (top 10%) |
| 2021      | First Prize, 13th Chinese College Mathematics Competition (Shanghai Division)      |

## SKILLS

|             |                                                                           |
|-------------|---------------------------------------------------------------------------|
| Programming | Python, C/C++, MATLAB, PyTorch, JAX, ROS, L <sup>A</sup> T <sub>E</sub> X |
| Languages   | English, Chinese (native)                                                 |